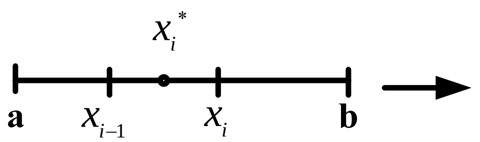
§5-2 The Definite Integral

Homework：1,9,19,23,29,33,35,37,41,47,50,53,67,68.

介紹定積分$\int_a^b f(x)dx$的定義，幾何意義和性質。



n等分,$\left[x_{i-1,}x_i\right]$=the ith subinterval.

- Riemanm sum = $\sum_{i=1}^{n} f\left(x_i^*\right)\Delta x \quad ,\Delta x=\dfrac{b-a}{n}$

  $\overset{若\ f(x)\geq 0}{=} \sum$小長方形面積

- Definite integral of f from a to b

  $=\int_a^b f\left(x\right)dx := \lim\limits_{n\to\infty}$ Riemanm sum

  $=\lim\limits_{n\to\infty}\sum_{\tau=1}^{n} f\left(x_i^*\right)\Delta x \overset{f(x)\geq 0}{=}$ the area

  under the curve $y=f\left(x\right)$ from a to b.

- $f\left(x\right)$ is called the integrand ; a is the lower limit ; b is the upper limit.

Theorem：If f is continuous on $\left[a,b\right]$.

$\Rightarrow \int_a^b f\left(x\right)dx$ is well-defined.

That is, the limit of the Riemanm sum is independent of the choice of x_i^*.

- Well-definedness of $\int_a^b f\left(x\right)dx$. (註:不等分的子區間也對)
- $\sum$:離散加法 $\int$:連續加法

Example 1：Evaluate the Riemanm sum for $f\left(x\right)=x^2$ with $n=6$,

x_i^*= left endpoints, $a=0,\ b=3$.

Solution： $\frac{1}{2}\left[0^2+\left(\frac{1}{2}\right)^2+1^2+\left(\frac{3}{2}\right)^2+2^2+\left(\frac{5}{2}\right)^2\right]=\frac{55}{8}$.

Example 2： Compute $\int_0^3 x^3-6x\,dx$ by definition.

Solution：

$$S_n=\sum_{i=1}^{n}\frac{3}{n}f\left(\frac{3i}{n}\right)=\sum_{i=1}^{n}\frac{3}{n}\left(\frac{27i^3}{n^3}-\frac{18i}{n}\right)$$

$$=\frac{81}{n^4}\sum_{i=1}^{n}i^3-\frac{54}{n^2}\sum_{i=1}^{n}i=\frac{81n^2(n+1)^2}{4n^4}-\frac{27n(n+1)}{n^2}$$

$$=\frac{81}{4}-27=\frac{-27}{4}.$$

Example 3： Evaluate $\int_0^1\sqrt{1-x^2}\,dx$ by interpreting in terms of area.

Sol：$\int_0^1\sqrt{1-x^2}\,dx=$ 四分之一的單位圓面積$=\frac{\pi}{4}$.

Example 4： $\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+\left(\frac{i}{n}\right)^2}=\int_{\square}^{\square}\square\,dx$

(Fill in the blanks, that is, identify f, a and b.)

Solution : $a=0$, $b=0$, $f(x)=\frac{1}{1+x^2}$.

(註：這種問題解答不是唯一)

Properties of the Integral

- $\int_a^b f(x)\,dx=-\int_b^a f(x)\,dx$.
- $\int_a^a f(x)\,dx=0$.
- $\int_a^b\left[\alpha f(x)+\beta g(x)\right]dx=\alpha\int_a^b f(x)\,dx+\beta\int_a^b g(x)\,dx$.

  (積分和微分一樣都是線性運算)
- $\int_a^c f(x)\,dx+\int_c^b f(x)\,dx=\int_a^b f(x)\,dx$, for any $c\in R$.
- $f(x)\ge 0$ on $[a,b]\Rightarrow\int_a^b f(x)\,dx\ge 0$.
- $f(x)\ge g(x)$ on $[a,b]\Rightarrow\int_a^b f(x)\,dx\ge\int_a^b g(x)\,dx$.

- If $m \le f(x) \le M$ for $a \le x \le b$, then $m(b-a) \le \int_a^b f(x)\,dx \le M(b-a)$.